

WASP-36b

R-2094

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140320 · created 2026-08-10T01:10:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456736.69676 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.68 +/- 1.57 [%]
Orbital Inclination (inc)	81.98 +/- 2.88
Airmass coefficient 1 (a1)	3.02 +/- 2.12
Airmass coefficient 2 (a2)	-0.9 +/- 1.21
Scatter in the residuals of the lightcurve fit is	71.7 %
Best Comparison Star	#3 - [513, 30]
Optimal Aperture	2.88
Optimal Annulus	5.75
Transit Duration (day)	0.06 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1368 (deviation 0.52σ)
Duration fit vs published	0.06 d vs 0.0758 d (deviation 0.48σ)
Mid-time O–C	-2.4 min
Depth SNR	3.1
Residual scatter	71.7 %

Light curve

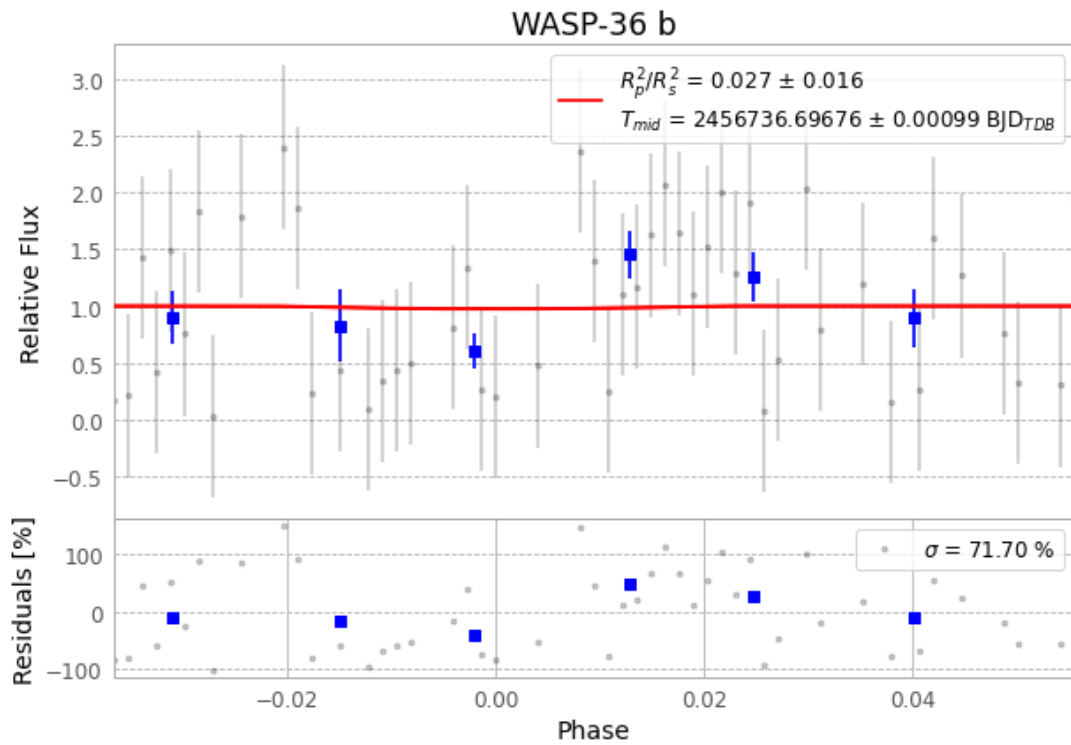


Figure 1 — FinalLightCurve_WASP-36 b_2014-03-19.png (SHA-256 92b4302d5a3a24c2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 414], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c49d8860ac31c3ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), WASP-36140320031212.FITS → WASP-36140320063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140320.log (19b4ecf2208c...), FinalLightCurve_WASP-36 b_2014-03-19.png (92b4302d5a3a...), FinalLightCurve_WASP-36 b_2014-03-19.csv (096651aa3523...)

Compute resources

Wall time 167.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 01:10Z.